

Course Code : ECT 452-2

GVHW/RW – 23 / 1028

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

(Program Elective – IV)

BROADBAND COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

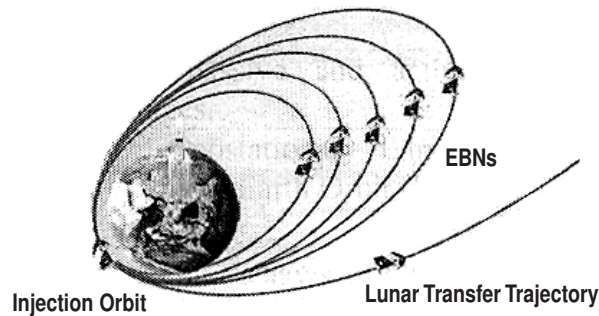
- (1) All questions are compulsory.
- (2) Assume suitable data wherever necessary.
- (3) All questions carry marks as indicated.
- (4) Draw neat diagrams wherever necessary.

1.
 - (a) Explain the types of ISDN. Differentiate between Basic rate Interface and Primary rate Interface in case of ISDN. 5(CO1)
 - (b) Compare cable broadband and fiber broadband technologies. 3(CO1)
 - (c) Explain with block diagram the use of SONET in B-ISDN. 2(CO1)
2.
 - (a) What are passive and active satellites ? State its advantages and disadvantages. 5(CO1)
 - (b) Explain :
 - (1) Geostationary Earth Orbit (GEO).
 - (2) Medium Earth Orbit (MEO).
 - (3) Low Earth Orbit (LEO).Along with their applications. 5(CO3)
3.
 - (a) Explain Kepler's laws of Planetary Motion. Illustrate in each case their relevance to artificial satellites orbiting the earth. 5(CO3)
 - (b) A Chandrayan-3 in second orbit raising activity has achieved 41603 km x 226 km elliptical orbit around the earth, while in its third orbit raising activity it has achieved 51400 km x 228 km orbit. What was the orbital period of

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Chandrayan-3 in second and third orbit ? Give your answer in hours.
 Note : Assume the average radius of the Earth is 6,378.137 km and Kepler's constant has the value $3.986004418 \times 10^5 \text{ km}^3/\text{s}^2$.



5(CO5)

4. (a) What are the various forces involved when satellite is revolving in orbit around Earth ? Derive the expression for Velocity of satellite in circular orbit and its time period. 5(CO3)
- (b) Explain the Attitude and Orbit control system for satellite in detail. 5(CO2)
5. (a) Draw a block diagram of a satellite transponder for 14/11 GHz band and explain its working. 5(CO2)
- (b) Explain with diagram the phenomenon of Earth's Eclipse of a Satellite and its effects on satellite communication. 5(CO4)
6. (a) A satellite at a distance of 36,000 km from building radiates a power of 20 W from an antenna with gain of 22 dB in direction of VSAT at building with effective aperture area 10 meter².
 Find :
 - (a) The flux density.
 - (b) Power received by VSAT antenna.
 - (c) If satellite operates at frequency of 11 GHz and Earth station antenna has gain of 52.3 dB. Determine received power. 6(CO5)
- (b) What are Uplink and Downlink ? Explain the Rain Effects on Downlink. 4(CO2)

